

SA Data Integration

Stokvel Management Platform | Software Design 2026

Overview

The project brief required each team to source and consume a real, publicly available South African dataset or API relevant to the project domain. This document describes our chosen data source, justification, implementation, and how it integrates into the platform.

Requirement

"The application must display the current South African prime lending rate and repo rate, calculated using live or regularly updated public data. Your team must research and identify a suitable publicly available South African data source for this information. Members' projected savings growth must be derived from this data."

Chosen Data Source

Organisation: South African Reserve Bank (SARB)

Source: SARB Selected Historical Statistics — Interest Rates

URL: <https://www.resbank.co.za/en/home/what-we-do/statistics/releases/selected-historical-statistics>

Data format: Publicly accessible via the SARB website and updated regularly following Monetary Policy Committee (MPC) decisions.

Justification

The South African Reserve Bank is the central bank of South Africa and the sole authoritative source for official monetary policy interest rates. We chose this source for the following reasons:

1. **Official and authoritative** — The SARB sets and publishes the repo rate directly. There is no more reliable source for this data in South Africa.
2. **Publicly available** — The data is freely accessible without authentication or API keys.

3. **Regularly updated** — The repo rate is updated following each MPC meeting, which occurs approximately every two months. The prime lending rate is derived directly from the repo rate (repo rate + 3.5 percentage points) and updates accordingly.
4. **Directly relevant** — Stokvels are savings groups. The prime lending rate and repo rate directly affect the return on savings and the cost of credit, which are core financial concepts for stokvel members.

Current Values (as at Sprint 4, May 2026)

Rate	Value
Repo Rate	6.75%
Prime Lending Rate	10.25%

Implementation

File: frontend/js/interest-rates.js

The interest-rates.js file handles fetching and displaying the live interest rate data. It also calculates projected savings growth for stokvel members based on the current prime rate.

How it works

Since the SARB does not provide a public REST API with JSON responses, the implementation uses one of the following approaches:

1. **Supabase database table** — The database teammate (Mashudu) created a `interest_rates` table in Supabase that is manually updated after each MPC announcement. The frontend fetches the latest rates from this table

Savings projection formula

The projected savings growth is calculated as follows:

$$\text{Projected Value} = \text{Contribution Amount} \times (1 + (\text{Prime Rate} / 100)) ^ \text{Years}$$

This gives members an estimate of how their stokvel savings could grow over time based on the current prime lending rate.

Where it appears in the application

The interest rate data is displayed in two places:

1. **Member Dashboard** — Shows the current repo rate and prime rate in a dedicated panel alongside the member's projected savings growth for their stokvel cycle.
2. **Admin/Treasurer Dashboard** — Displays the rates as context for financial planning and group management decisions.